Linuxzoo.net

Minimize

User: vinay mummadi (FULL)
Account: 40807459@live.napier.ac.uk
Registered Account
Account Links
Machine State: **RUN**
Refresh in: 0:20.
Time left: 50 mins
Boot progress: complete

control | connect | stats | useful

Switch On | Nice Switch Off
Pull Out Power | Leave Queue

Kali 2024-4
- fresh linux install image
- use previous image (if available)

Forged Emails

In this short tutorial we will forge some emails and analyse them

To reset all the check buttons from a previous attempt click here

Question 1 Email Server Running

Before you start you must be running a mail server. In Kali, the server to use is exim4. Start this service by typing

```
sudo systemctl start exim4
```

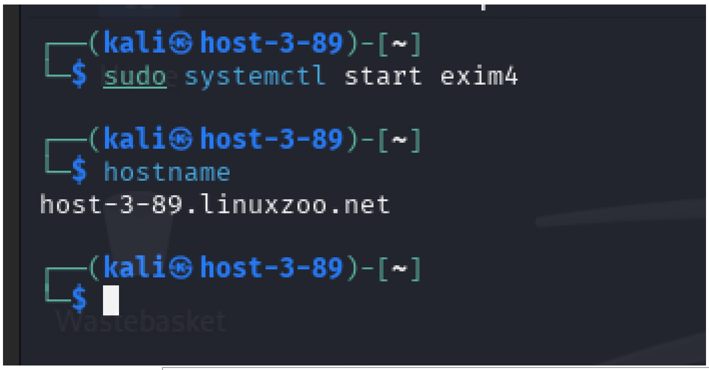
Check Tests: Ready

Service running PASS

Question 2 Local delivery

You are sending emails to ourselves. Find out what you hostname is by running "hostname".

What is your hostname?



hostname: host-3-89.linuxzoo.net

Check Tests: Ready

Hostname is right PASS

Question 3 netcat line

We are going to forge an email to ourselves.

Using the normal IP number of localhost, what would be the netcat command needed to netcat to the SMTP port of localhost? This is case-sensitive and space-sensitive. Use only numerical IP and port, and start the answer "netcat".

Netcat command: netcat 127.0.0.1 25

Check Tests: Ready

Netcat Command PASS

```
┌──(kali㉿host-3-89)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.3.89  netmask 255.255.255.248  broadcast 10.0.3.95
        inet6 fe80::20c:59ff:fe00:390  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:59:00:03:90  txqueuelen 1000  (Ethernet)
        RX packets 59  bytes 7203 (7.0 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 86  bytes 12957 (12.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

labbr0: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.1.254  netmask 255.255.255.0  broadcast 192.168.1.255
        ether 52:54:00:a6:b7:32  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 6 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 12  bytes 786 (786.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 12  bytes 786 (786.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0


┌──(kali㉿host-3-89)-[~]
└─$ netcat 127.0.0.1 25
220 host-3-89.linuxzoo.net ESMTP Exim 4.98 Fri, 27 Feb 2026 13:12:46 +0000
QUIT
221 host-3-89.linuxzoo.net closing connection
```

Question 4 Use the netcat command

Use this netcat command (and use QUIT in the netcat session) to test the connection. The MTA (Mail Transport Agent) server should identify itself with a string "ESMTP MTANAME Version". What is the version of the MTA?

MTA version: 4.98

Check Tests: Ready

MTA version PASS

Question 5 Envelope Information

We want to forge an email and make it appear that: (1) The sending server is "juggling.com", (2) the email is coming from "dave@rocketman.com", and (3) the email is going to "kali@YOURHOST" (where YOURHOST is the hostname you have already discovered for your own virtual machine).

So what are those fields in this case?

Type first? (hint - the helo):

Type second? (hint - from who):

Type third? (hint - to who):

Remember to include the entire line you would type in SMTP. Case and space sensitive, so always use lowercase in answering.

```
┌──(kali㉿host-3-89)-[~]
└─$ netcat 127.0.0.1 25
220 host-3-89.linuxzoo.net ESMTP Exim 4.98 Fri, 27 Feb 2026 14:29:48 +0000
helo juggling.com
250 host-3-89.linuxzoo.net Hello juggling.com [127.0.0.1]
mail from: dave@rocketman.com
250 OK
rcpt to: kali@host-3-89.linuxzoo.net
250 Accepted
data
354 Enter message, ending with "." on a line by itself
subject: hi
.
250 OK id=1vvyrk-00000000LIM-2McR
QUIT
221 host-3-89.linuxzoo.net closing connection

┌──(kali㉿host-3-89)-[~]
└─$ 
```

Check Tests: Ready

HELO	UNTESTED
from	UNTESTED
to	UNTESTED

Question 6 Forge an email

Use the above information to send an email. Open the connection with netcat as above, use all the header data as the previous question, then in the data body section send the following:

```
From: me@thegovernment.com
To: you@thepeople.com
Date: today
Subject: stupid

Please send me your bank details.
Boss
.
```

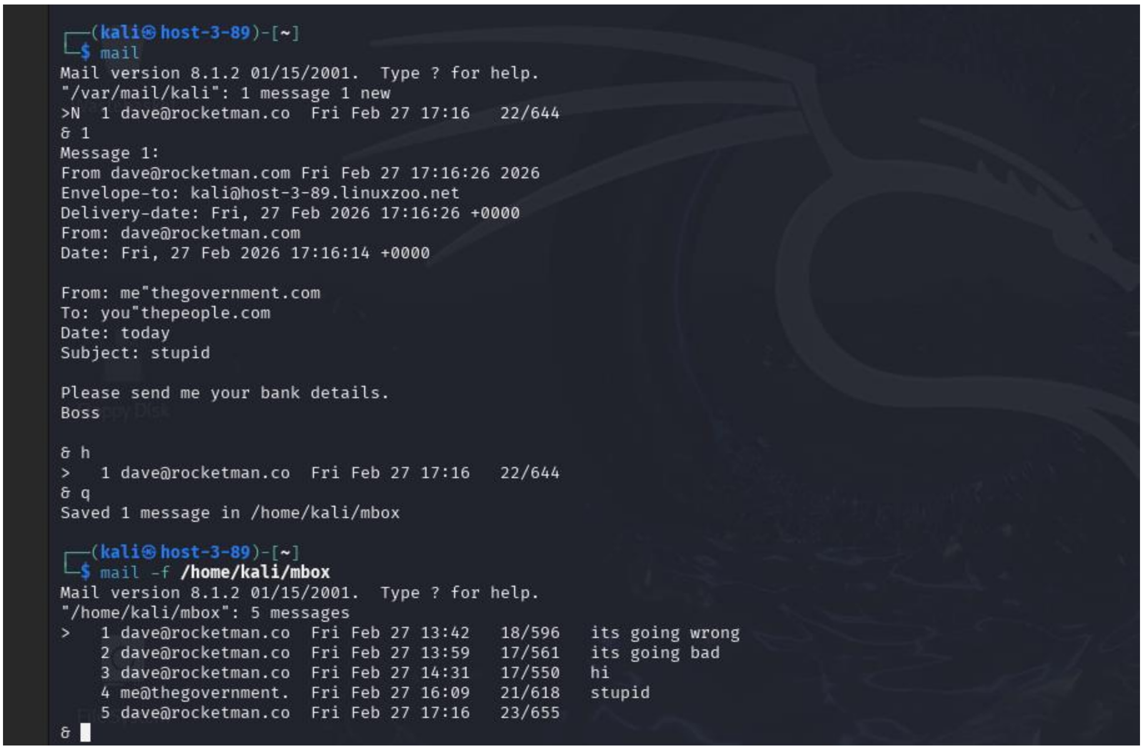
```
┌──(kali㉿host-3-89)-[~]
└─$ netcat 127.0.0.1 25
220 host-3-89.linuxzoo.net ESMTP Exim 4.98 Fri, 27 Feb 2026 17:14:24 +0000
helo juggling.com
250 host-3-89.linuxzoo.net Hello juggling.com [127.0.0.1]
mail from: dave@rocketman.com
250 OK
rcpt to: kali@host-3-89.linuxzoo.net
250 Accepted
data
354 Enter message, ending with "." on a line by itself

From: me"thegovernment.com
To: you"thepeople.com
Date: today
Subject: stupid

Please send me your bank details.
Boss
.
250 OK id=1vw1QM-000000001Uw-0NFH
quit
221 host-3-89.linuxzoo.net closing connection
```

After the body remember to end the block with ".", then end the session with "quit".

To read the mail for "mail", do:

```
Mail
```

To delete mail, do "d 1" where "1" is the number of the mail to delete. Do "1" to read email number 1. Do "h" to see the emails which can be written. Press "q" to actually delete email "It does not get deleted till you q out of mail".



```
┌──(kali㉿host-3-89)-[~]
└─$ mail
Mail version 8.1.2 01/15/2001.  Type ? for help.
"/var/mail/kali": 1 message 1 new
>N  1 dave@rocketman.co  Fri Feb 27 17:16   22/644
& 1
Message 1:
From dave@rocketman.com Fri Feb 27 17:16:26 2026
Envelope-to: kali@host-3-89.linuxzoo.net
Delivery-date: Fri, 27 Feb 2026 17:16:26 +0000
From: dave@rocketman.com
Date: Fri, 27 Feb 2026 17:16:14 +0000

From: me"thegovernment.com
To: you"thepeople.com
Date: today
Subject: stupid

Please send me your bank details.
Boss

& h
>   1 dave@rocketman.co  Fri Feb 27 17:16   22/644
& q
Saved 1 message in /home/kali/mbox

┌──(kali㉿host-3-89)-[~]
└─$ mail -f /home/kali/mbox
Mail version 8.1.2 01/15/2001.  Type ? for help.
"/home/kali/mbox": 5 messages
>   1 dave@rocketman.co  Fri Feb 27 13:42   18/596   its going wrong
    2 dave@rocketman.co  Fri Feb 27 13:59   17/561   its going bad
    3 dave@rocketman.co  Fri Feb 27 14:31   17/550   hi
    4 me@thegovernment.  Fri Feb 27 16:09   21/618   stupid
    5 dave@rocketman.co  Fri Feb 27 17:16   23/655
&
```

Note: make sure that their is only 1 email in the mailbox and that the one which is there is the one you are forging. Remember to "q" out of mail.

Note if you read the email with "mail" then quit, it moves it to /home/kali/inbox. If it does and then you want to look there, then do "mail -f /home/kali/mbox". The buttons look in both places.

Check Tests: Ready

Email envelope detected in log	UNTESTED
Email path includes juggling.com	UNTESTED
BODY From is ok	UNTESTED
BODY To is ok	UNTESTED

Question 7 Hidden Path

Run the command:

```
grep -A 5 "Received:" /var/spool/mail/kali
```

This gives the line with the "Received:" on it, as well as the next 2 lines. This shows the mail path. Here it is clear that "juggling.com" would have an IP number, but the IP number in the path (on line 1 in the square brackets) is the IP of localhost (in ipv6 addressing) and not of juggling.com. Thus it is forged. In linux the dig command will give you the IP.

```
┌──(kali㉿host-3-89)-[~]
└─$ dig juggling.com

; <<>> DiG 9.20.4-3-Debian <<>> juggling.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 41839
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: d0adba90cfe6cba50100000069a1d6e310e75a9905ca5d14 (good)
;; QUESTION SECTION:
;juggling.com.                  IN      A

;; ANSWER SECTION:
juggling.com.           86400   IN      A       10.255.5.4

;; Query time: 0 msec
;; SERVER: 10.200.0.1#53(10.200.0.1) (UDP)
;; WHEN: Fri Feb 27 17:39:46 GMT 2026
;; MSG SIZE  rcvd: 85
```

Validate this by entering the IP number of juggling.com IP of juggling.com?

Check Tests: Ready

IP of juggling.com	UNTESTED

Question 8 Better Forgery

Repeat the forged email exercise, but this time include one fake hop at the head of the data section. Use the grep information to make the hop identical to the one created when you sent the first email, except this time replace the sending IP with the juggling.com IP and replace the hostname "localhost" with "juggling.com".

The check buttons look for the newest email in the inbox, and if that is empty the newest one in mbox

So if you assume the header was:

```
Received: from localhost ([::1] helo=juggling.com)
        by host-19-17.linuxzoo.net with smtp (Exim 4.80)
        (envelope-from <dave@rocketman.com>)
        id 1WIgrz-00021U-Ge
        for root@host-19-17.linuxzoo.net; Wed, 26 Feb 2014 15:58:09 +0000
```

Then use this whole entry, but replace [::1] with [fakeIP] (i.e. the ip of the real juggling.com) and "localhost" with "juggling.com".

```
┌──(kali㉿host-3-89)-[~]
└─$ netcat 127.0.0.1 25
220 host-3-89.linuxzoo.net ESMTP Exim 4.98 Fri, 27 Feb 2026 18:08:50 +0000
helo juggling.com
250 host-3-89.linuxzoo.net Hello juggling.com [127.0.0.1]
mail from: dave@rocketman.com
250 OK
rcpt to: kali@host-3-89.linuxzoo.net
250 Accepted
data
354 Enter message, ending with "." on a line by itself
Received: from juggling.com ([10.255.5.4] helo=juggling.com)
        by host-19-17.linuxzoo.net with smtp (Exim 4.80)
        (envelope-from <dave"rocketman.com>)
        id 1WIgrz-00021U-Ge
        for root"host-19-17.linuxzoo.net; Wed, 26 Feb 2014 15:58:09 +0000


From: me@thegovernment.com
To: you@thepeople.com
Date: today
Subject: stupid

Please send me your bank details.
Boss
.
250 OK id=1vw2Hm-000000008S8-2beE
quit
221 host-3-89.linuxzoo.net closing connection
```

Check Tests: Ready

Email envelope detected in log UNTESTED

Email path includes juggling.com	UNTESTED
BODY From is ok	UNTESTED
BODY To is ok	UNTESTED
2 Received hops in email	UNTESTED
Fake hop looks good	UNTESTED